

Imperial College London

MSc EXAMINATION June 2023

This paper is also taken for the relevant Examination for the Associateship

MRES - STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS

For MLBD MRes Students

June 5th, 2023: 10:00 to 12:00

Answer all FOUR questions

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. The Maxwell-Boltzmann distribution describes the distribution of the speed, $v \in [0, +\infty)$, of molecules in a gas at temperature T . Its probability density is given by,

$$f(v; T) = \left(\frac{m}{2\pi kT} \right)^{3/2} 4\pi v^2 e^{-\frac{mv^2}{2kT}},$$

where m is the molecular mass and k is a constant known as the Boltzmann constant. The speed v of any molecule will be a random variable $v \sim f(v; T)$.

- (i) Show that the first algebraic moment (the expectation of v) of the Maxwell-Boltzmann distribution μ_1 is given by,

$$\mu_1 = E[v] = 2 \sqrt{\frac{2kT}{m\pi}}.$$

You may use the following identity,

$$\int_0^{+\infty} x^3 e^{-bx^2} dx = \frac{1}{2b^2}.$$

[3 marks]

- (ii) Imagine an experimental apparatus designed to determine the speeds of molecules in a gas v . This is possible at least in terms of measuring small intervals of speeds but don't worry about the details. Suppose the experiment observes a large number n of speeds $\{v_1, v_2, \dots, v_n\}$. These observations are independent and identically distributed random variables distributed as $v_i \sim f(v; T)$.

- (a) Show that the sample mean, $\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$, is an unbiased estimator for the first algebraic moment of the the Maxwell-Boltzmann distribution. [4 marks]
- (b) Using the method of moments, show that an estimator for the temperature of the gas $\hat{T}$ based on the sample mean, is given by,

$$\hat{T} = \frac{\pi m \bar{v}^2}{8k}.$$

[1 mark]

- (c) Is the estimator $\hat{T}$ a biased or unbiased estimator of T ? Hint: Use the fact that,

$$\bar{v}^2 = (\bar{v} - \mu_1)^2 + 2\mu_1(\bar{v} - \mu_1) + \mu_1^2,$$

and assume that the second central moment (the variance) of the Maxwell-Boltzmann distribution, $v_2 = E[(v - \mu_1)^2]$, is non zero. [8 marks]

- (d) A second estimator, $\hat{T}'$ can be defined based on the second algebraic moment $\mu_2 = \frac{3kT}{m}$ and the second algebraic sample moment for the data, $m_2 = \frac{1}{n} \sum v_i^2$. By equating μ_2 and m_2 , find the expression for $\hat{T}'$ and show that this estimator is unbiased. [6 marks]

- (iii) Name one other property, aside from being unbiased, of a *good* estimator and describe what that property means. You can use equations and/or diagrams in your answer. [3 marks]

[Total 25 marks]

2. Hypothesis testing is the frequentist process of rejecting (or accepting) a scientific hypothesis given some observation in experimental data. Formally two hypotheses are defined, $H_0 :=$ the null hypothesis, and $H_1 :=$ the alternate hypothesis.

(i) (a) State the definition of Type-I and Type-II errors in terms of the two hypotheses H_0 and H_1 . [3 marks]

(b) A testing apparatus is designed to detect for a faulty piece of lab equipment. The test will report ‘N’ if the equipment is faulty and ‘T’ if the equipment is fine. The designer claims that the probability that the test reads ‘N’ when equipment is faulty is 75%. You test out a piece of equipment in your lab with the apparatus and get a reading ‘N’. Which of the following statements is true given only the information you have in the question.

A The probability that the equipment is faulty is 75%.

B If the equipment is not faulty, the probability that the test yields ‘N’ is 25%.

C The probability that the test would have read ‘T’ if the equipment was faulty is 25%

[2 marks]

(ii) An experiment is designed to detect decays of muon particles after being produced in an accelerator. A detector is setup to record the decay times, t , after the muon is produced at $t_0 = 0$. The experiment runs until a maximum time T such that no decay times $t > T$ are recorded. The probability to observe a decay time t after is given by the exponential probability distribution,

$$f(t; \tau) = N \exp\left(-\frac{t}{\tau}\right)$$

where τ is the *lifetime* parameter.

(a) Derive the normalisation constant N in terms of τ and T . [3 marks]

(b) The figure below (Figure 1) shows a distribution in data (observed values of the decay time t) collected by the experiment, and the exponential probability distribution.

Describe how you would calculate the cumulative empirical distribution function $F_{emp}(t)$ for the data. [2 marks]

(c) The Kolomogorov-Smirnov test-statistic is given by,

$$D = \max \left\{ |F_{emp}(t) - F(t)| \right\},$$

where $F(t)$ is the cumulative distribution of the exponential function in this case. Derive $F(t)$. [3 marks]

(d) A test is performed to see if the exponential is a good description for the data shown. The figure below (Figure 2) shows the distribution of D assuming the exponential shown is the true distribution ($f(D|H_0)$) and the observed value of the test statistic D_{obs} . The figure also shows a value D_{95} for which 95% of the distribution is to the left of the line. Describe how the information in the figure could be used to test if the hypothesis that the exponential is a good description of the data is true or not. Based on this, do you think the exponential is a good description of the data or not? [2 marks]

[This question continues on the next page ...]

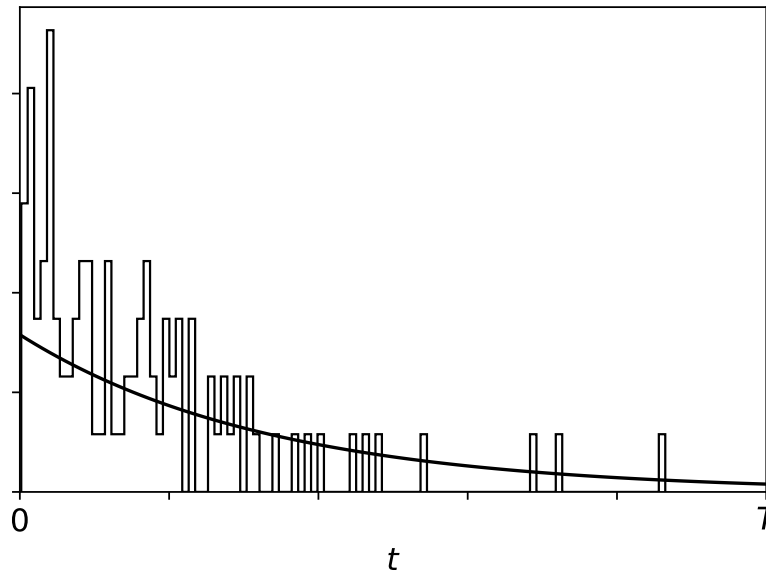


Figure 1

- (iii) With the experimental data from part (ii), it should be possible to measure the parameter τ . Assume for the rest of this question that the times observed are much smaller than the maximum time T or equally that $T \rightarrow +\infty$, and that the observed times are independent and known precisely.
- Write down the likelihood function for the set of n observations $\{t_1, t_2, \dots, t_n\}$ assuming the same exponential probability distribution from part (ii).
[2 marks]
 - Derive the maximum likelihood estimate of the parameter $\hat{\tau}$ and express the answer in terms of a sample moment of the dataset.
[3 marks]
 - Explain how you would construct a 68% confidence interval for the parameter τ , using the Neyman construction. You should clearly state all of the steps required and you can write your answer in terms of pseudo-code if you wish.
[5 marks]

[Total 25 marks]

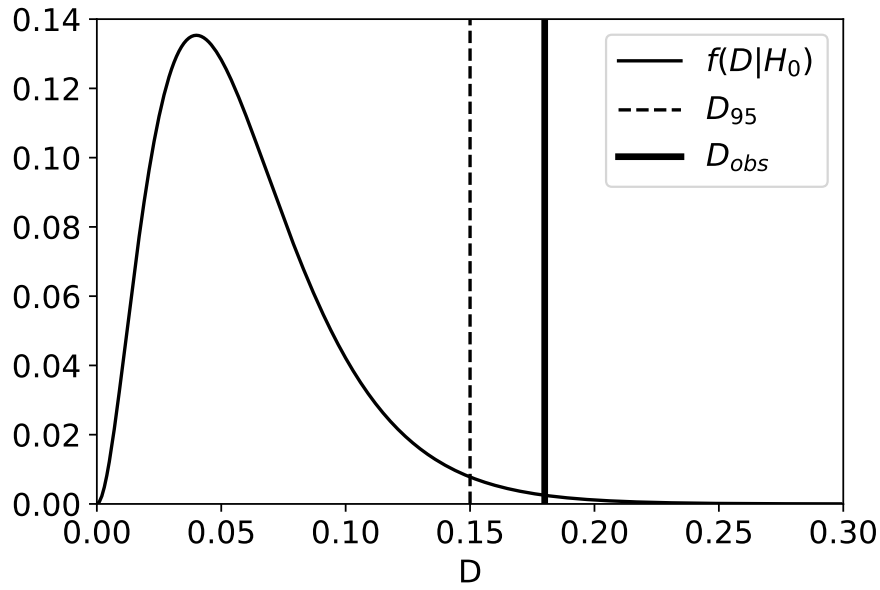


Figure 2

3. Two sets of data, $\mathbf{d}_1$ and $\mathbf{d}_2$ are collected, and used to infer parameters θ of a model. In a Bayesian analysis, where the prior state of knowledge before either set of data is collected is $\pi(\theta)$.

- (i) Find an expression for the final posterior $p(\theta|\mathbf{d}_1, \mathbf{d}_2)$ computed by a two-step process, first updating the original prior $\pi(\theta)$ to a posterior given $\mathbf{d}_1$, $p(\theta|\mathbf{d}_1)$, and using this as a prior for the analysis of $\mathbf{d}_2$. and show that this gives the same result as updating our state of knowledge from $\pi(\theta)$ with the information from the combined dataset $\mathbf{d}_1, \mathbf{d}_2$. i.e. show that the two-step analysis yields the expected

$$p(\theta|\mathbf{d}_2, \mathbf{d}_1) = \frac{p(\mathbf{d}_2, \mathbf{d}_1|\theta)\pi(\theta)}{p(\mathbf{d}_2, \mathbf{d}_1)}.$$

[7 marks]

- (ii) A physics theory makes a prediction for the probability q that an experiment detects a particle. An experiment is performed N times, and M independent detections are made.

- (a) Argue that the probability of obtaining this outcome is the binomial distribution,

$$p(M|N, q) = \binom{N}{M} q^M (1-q)^{N-M}.$$

- (b) A conjugate prior for the binomial distribution is the beta distribution,

$$\pi(q) = A(\alpha, \beta) q^{\alpha-1} (1-q)^{\beta-1}$$

where α and β are hyperparameters of the prior, and A serves to normalise the probability distribution. What is a conjugate prior?

[This question continues on the next page ...]

- (c) Describe qualitatively what prior knowledge is reflected in values $(\alpha, \beta) = (1, 1), (2, 1/4), (2, 2)$.

[6 marks]

- (iii) (a) Assuming a beta distribution prior, find an expression for the posterior for the experiment above.
- (b) Show that if two sets of trials are undertaken, yielding M_1 detections from N_1 trials, and then a further M_2 detections from N_2 trials, find explicit expressions for the final posterior (up to a normalising constant that you do not need to consider) and show that they are the same whether the sets are analysed in sequence, or the data combined first and analysed only once. Use the beta prior as before.

[6 marks]

- (iv) (a) Find the MAP (maximum a posteriori) value of q , for given α and β . If $\alpha + \beta = 2$, comment on how it depends on α , as well as on the data. You may wish to simplify the algebra by defining $C = M + \alpha - 1$ and $D = N - M + \beta - 1$.
- (b) By expanding the log(posterior) around the peak value, show that the uncertainty on q may be written as

$$\sigma_q^{-2} = \frac{(N + \alpha + \beta - 2)^3}{(N - M + \beta - 1)(M + \alpha - 1)}.$$

[6 marks]

[Total 25 marks]

4. The decay rate $\mu > 0$ for a radioactive process (whose signal is proportional to $\exp(-\mu t)$ for $t \geq 0$) is to be inferred from independent measurements, denoted $\hat{t}_i$ ($i = 1, \dots, N$), of decay times. These measured values are known to be imprecise, with uncorrelated gaussian errors of known variances σ_i^2 .

- (i) (a) What are latent variables and which ones occur in this problem?
 (b) If t is the time of a decay, write down a properly normalised expression for the sampling distribution of t in terms of the decay rate parameter μ .

[5 marks]

- (ii) (a) Draw a directed acyclic graph (DAG) that represents the process, identifying quantities that are drawn from probability distributions with open circles, known quantities with a dot, deterministic quantities in a diamond, and quantities drawn from the same distribution should be surrounded with a rectangular 'plate', identifying the number of repetitions. Measured data should be in shaded circles. Include any latent variables identified in the previous part.

[6 marks]

- (iii) (a) For a given prior $\pi(\mu)$, write down an expression for the posterior for μ using a Bayesian hierarchical model based on the DAG. Make use of the rules of probability and marginalise over the latent variables, to write your answer in terms of a product of integrals.

[8 marks]

- (iv) Prove that the posterior is given by

$$p(\mu | \{\hat{t}_i\}) \propto \mu^{N-1} \prod_{i=1}^N \exp\left(\mu \hat{t}_i - \frac{\mu^2 \sigma_i^2}{2}\right) \left[1 + \operatorname{erf}\left(\frac{\hat{t}_i}{\sqrt{2}\sigma_i} - \frac{\mu\sigma_i}{\sqrt{2}}\right)\right]$$

for a particular choice of prior, and justify that prior. Use the gaussian integral given at the end of the question and explicitly write down expressions for a , b , and c .

[6 marks]

You may assume without proof the result

$$\int_0^\infty \exp(-ax^2 + bx + c) dx = \sqrt{\frac{\pi}{4a}} \exp\left(\frac{b^2}{4a} + c\right) \left[1 + \operatorname{erf}\left(\frac{b}{2\sqrt{a}}\right)\right].$$

[Total 25 marks]